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A chemical kinetic study of the oxidation of silane by nitrous oxide, nitric oxide and oxygen

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Abstract

Silane–nitrous oxide mixtures are widely used in the semi-conductor manufacturing industry. Although these mixtures are potentially hazardous, their explosive properties are not very well characterised. In the present study, the high temperature kinetics of highly argon diluted silane–nitrous oxide mixtures has been examined. The dynamics of oxygen atoms behind reflected shock waves has been studied by Atomic Resonance Absorption Spectroscopy at temperatures between 1725 and 2584 K and pressures from 231 to 601 kPa. In order to take into account the effect of silane molecule and silicon atom absorption at 130.5 nm, absorption cross section determinations have been performed. In our experimental conditions (compositions and observation times), steady state oxygen atom concentrations have been observed at low temperatures, after few hundreds of microseconds. At high temperatures, the initial fast increase of oxygen atom concentration was followed by a period of consumption. A kinetic model that reproduces very well the experimental absorption profiles has been developed. Important reactions have been underlined through sensitivity and reaction pathway analyses. Previous experimental results from the literature for the oxidation of silane by nitrous oxide, nitric oxide and oxygen behind reflected shock waves have also been compared to the results coming from the kinetic model.
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Keywords: Silane; Nitrous oxide; Reflected shock wave; Atomic Resonance Absorption Spectroscopy

1. Introduction

Silane–nitrous oxide mixtures are widely used in the semi-conductor industry to produce isolant or protective solid layers [1,2]. Because of silane high reactivity and pyrophoric behaviour, numerous incidents occurred in this industry [3]. Concerning silane–nitrous oxide mixtures, Hirano [4] reported an explosion in Japan which was attributed to this mixture. Although such blends are potentially hazardous, their explosive properties are not very well characterised. Horiguchi et al. [5] studied silane–nitrogen oxide mixtures flammability limits. For SiH_4–N_2O mixtures, ignition was possible from 1.90 to 87.1 mol% of silane, at room temperature and atmospheric pressure. Some studies have been realised on the kinetics of this system. Chapple-Sokol et al. [1,2] proposed a detailed kinetic model based on chemical vapour deposition experiments in a laminar flow reactor.

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The temperature of the substrate was between 773 and 923 K and the pressure in the reactor was atmospheric. The following parameters were measured to derive the kinetic scheme: the SiO_2 film growth peak amplitude and position, their evolution as a function of temperature and composition, silane conversion efficiency and solid film stoichiometry. Votintsev et al. [6] studied the reaction of silylene with nitrous oxide, $SiH_2 + N_2O = H_2SiO + N_2$, between 1400 and 1800 K and at pressure from 70 to 90 kPa. The rate constant of this reaction was obtained from the chemiluminescence signal of H_2SiO^*. The $SiH_2 + N_2O = H_2SiO + N_2$ reaction was also studied by Becerra et al. [7] at temperatures between 295 and 747 K and at pressures from 0.7 to 7 kPa. SiH_2 radicals were produced by flash photolysis of phenylsilane at 193 nm and their evolution was monitored by laser absorption at 579 nm. Mick and Roth [8] studied Si- and N-atom dynamics behind reflected shock waves by Atomic Resonance Absorption Spectroscopy (ARAS). The temperature was between 1780 and 3560 K and the pressure ranged from 50 to 170 kPa. Based on these measurements, the rate constants of the reactions: $Si + N_2O = SiO + N_2$ and $Si + N_2O = SiN + NO$ were obtained.

The bibliographic review shows that the available data on the silane–nitrous oxide mixtures are very limited. The present work aims at obtaining new experimental data for this system. The dynamics of oxygen atoms have been studied experimentally by ARAS and the results are interpreted on the basis of a detailed kinetic scheme.

First, the materials and the methods used are described. Second, the experimental and numerical results obtained are presented and discussed. Results from the kinetic scheme are compared to previous experimental results from the literature for the oxidation of silane by nitrous oxide, nitric oxide and oxygen behind reflected shock waves.

2. Materials and methods

2.1. Mixtures preparation

All gases used were supplied by Air Liquide. Research grade nitrous oxide, argon and a mixture of 1% of silane in argon were employed. The partial pressure method was used to prepare the mixtures in an apparatus evacuated to less than 10^{-5} Pa with a turbo-molecular pump. The silane content was set at 10 ppmv and the nitrous oxide content ranged from 10 to 40 ppmv.

2.2. Shock tube and optical arrangement

The shock tube used has been described in previous papers [9–11]. Briefly, the apparatus was a 78 mm internal diameter stainless-steel shock tube with a 5.5-m-long driven section and a 3.5-m-long driver section. A turbo-molecular pump evacuated the test section to less than 2.10^{-4} Pa, and the typical leak-plus-outgassing rate was of the order of 2.10^{-6} Pa/s or less. Helium was used as driver gas, and the shock wave was initiated by the bursting of a double diaphragm. The shock wave velocity was monitored by piezoelectric pressure gauges mounted flush to the inner wall. Thermodynamic conditions behind the incident and reflected shocks were computed from the experimental incident shock velocity using SHOCK [12].

The optical detection technique for measuring O-atom concentration was an emission line absorption method. A microwave-excited discharge lamp that contained a flowing mixture of 1% O_2 in He maintained at a pressure of 1.4 kPa was used as light source. A vacuum ultraviolet (UV) monochromator was used to isolate the O-triplet wavelength, and a solar blind photomultiplier was used to convert vacuum UV photons. The absorption measurements were made behind the reflected shock through two MgF_2 windows located at 10 mm from the end wall. The O-atom calibration was obtained from nitrous oxide pyrolysis experiments. Mixtures containing 2.9–29.3 ppmv of N_2O in Ar were shock-heated in the temperature range 1900–3750 K and for pressure between 170 and 310 kPa in order to obtain steady-state O concentrations. The following modified Lambert–Beer law was obtained:

$$A_O = 1 - \exp(-3.38 \times 10^{-10} \times \ell \times [O]^{0.625})$$

where A_O is the absorption by O atoms, ℓ the length of the optical path (7.8 cm) and [O] the O-atom concentration expressed in atoms cm^{-3}.

Our optical set-up allowed the measurement of O-atom concentration ranging from 1.10^{13} to 1.10^{14} atom cm^{-3}.

2.3. Chemical kinetic scheme

The detailed kinetic model used was developed using previously published schemes and some kinetic studies. The system H_2–N_2O was described by the model of Mével et al. [13]. Three changes were brought to this sub-set: (i) the rate coefficient for the $O + H_2 = OH + H$ rate was that from [9], (ii) the reaction rate for the $NO + N \rightarrow N_2 + O$ reaction was that from [14], (iii) the reaction $N + OH = NO + H$ was added and taken from [8]. The silane pyrolysis sub-set came from [15]. The sub-mechanism for the kinetics of the Si–H–O system was constructed from the studies of Miller et al. [16], Babushok et al. [17], Zachariah and Tsang [18] and Mick [19]. The interactions between Si and SiH_2 and the N_yO_x species came from [7,8,20]. The complete mechanism was composed of 448 elementary reactions and 92 chemical species and is available as a supplemental data.

The thermodynamic data came from [13,15,16]. None of the rate coefficients was adjusted.

The modeling of the experimental results was performed by using the SENKIN code [21] with the constant volume reactor model. First order sensitivity analyses were performed using this code. Sensitivity coefficients have been normalized with respect to the highest coefficient. Reaction pathway analyses were also performed using rates of production and destruction.

3. Results and discussion

3.1. Absorption cross sections of SiH_4 and Si

In the present study, some silane pyrolysis experiments have been performed (Table 1). It was found that both silane molecules and silicon atoms absorb at the O-triplet emission line at 130.5 nm. One can see in Fig. 1: (i) prior to the incident shock, the absorption of SiH_4 at ambient temperature, (ii) between the incident and the reflected shocks, the absorption of SiH_4 at incident shock temperature, T_2, and (iii) after the reflected shock, the absorption of Si atoms at reflected shock temperature, T_5. Indeed, on one hand, T_2 is too low to initiate the decomposition of SiH_4 and at T_5, SiH_4 decomposition is very fast and leads rapidly to a constant concentration of Si atoms, as shown in Fig. 1. To take into account the contribution of the absorption of theses species to the total absorption monitored during the experiments with SiH_4–N_2O–Ar mixtures, the absorption cross sections of SiH_4 and of Si atoms at 130.5 nm have been determined. Figure 2 gives a comparison between the evolution of the measured absorption by silane molecules as a function of silane concentration and those calculated from the absorption cross sections found in the literature [22–24]. The main source of uncertainties for the determination of the absorption by SiH_4 was the noise of the absorption signal. Whatever the absorption value was, the absolute uncertainty was of ±5%. Considering this uncertainty, the absorption cross section presently obtained, $\sigma_{SiH4} = 6.91 \times 10^{-17}$ cm^2 molecule^{-1}, is in very good agreement with those determined by [22–24], ranging between 6.4×10^{-17} and 7.7×10^{-17} cm^2 molecule^{-1}. Concerning the Si atoms, experimentally it was found $\sigma_{Si} = 5.62 \times 10^{-17}$ cm^2 atom^{-1}. Although no data

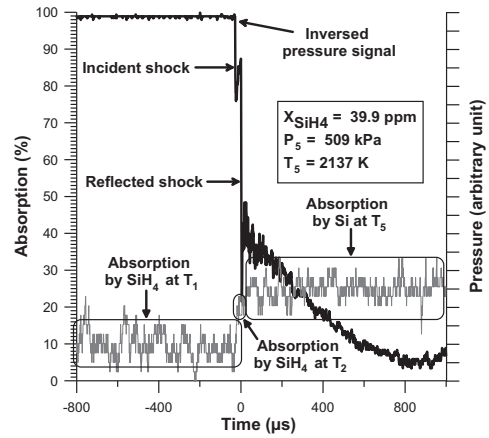

Fig. 1. Absorption by SiH_4 and Si at 130.5 nm for a SiH_4–Ar mixture (grey line) and inversed pressure signal (black line).

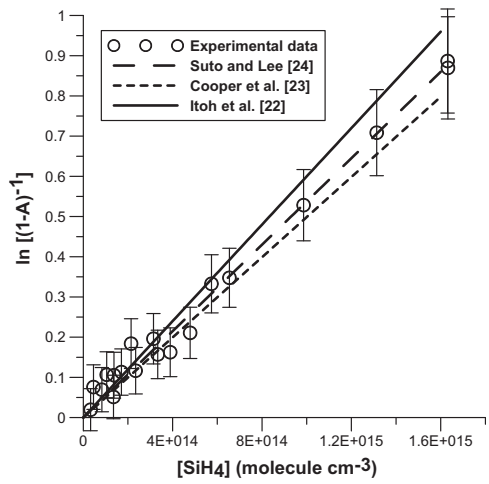

Fig. 2. Absorption by SiH_4 at 130.5 nm as a function of SiH_4 concentration. Comparison with calculated data obtained with absorption cross sections taken from references [22–24]. Experimental conditions: $X_{SiH4} = 10$–200 ppmv; $T_{1-2} = 294$–1135 K; $P_{1-2} = 4.7$–196 kPa.

Table 1
Composition and experimental conditions for the mixtures presently studied. In all cases, the diluent is argon.

Studied parameters	Mixture compositions		Experimental conditions		Absorption technique at:	References
	X_{SiH4} (ppm)	$X_{Oxidant}$ (ppm)	T (K)	P (kPa)		
A_{SiH_4}, A_{Si}	10–200	–	294–2257	4.7–679	SiH_4, Si: 130.5 nm	Present study
A_O	9.9–10.3	N_2O: 10–39.8	1725–2584	231–601	O: 130.5 nm	Present study
[Si], [N]	0.5–15	N_2O: 30–200	1922–2810	65–120	Si: 251.6 nm/N: 119.9 nm	[8]
A_{Si}, [N], [O]	0.75–10	NO: 300	2500–2955	50–110	Si: 251.6 nm/N: 119.9 nm/ O: 130.5 nm	[20]
[Si], [O]	1–30	O_2: 10–30	2300–2500	85–125	Si: 251.6 nm/O: 130.5 nm	[19]

have been found in the literature for comparison, this value seems consistent with that measured for SiH_4. Both absorption cross sections, for SiH_4 and Si, have been found independent of the temperature within the experimental conditions. As for the previous study carried out with O-ARAS [11], the contribution of N_2O to the total absorption has been taken into account.

3.2. Oxidation of silane by nitrous oxide

The absorption by O atoms at 130.5 nm has been measured in SiH_4–N_2O–Ar mixtures at reflected shock temperatures between 1725 and 2598 K and pressures from 231 to 601 kPa. Table 1 summarises the experimental conditions. Figures 3–5 present some typical results obtained. Since there were some contributions to the absorption by SiH_4, Si, and N_2O, absorption profiles are shown rather than O concentration profiles. At low temperatures (Fig. 3), from 1700 to 1850 K, a relatively fast increase of the absorption was observed. After about 1 ms, the absorption reached a quasi steady-state level up to 1800 μs the observation time at such temperatures. At intermediate temperatures (Fig. 4), around 2000 K, a faster increase of the absorption can be seen. After about 200 μs, a smooth decrease of the absorption level was observed. At high temperatures (Fig. 5), around 2500 K, the very fast increase of the absorption level was followed by a rapid decrease. The nonzero absorption at the initial time can be explained by the contributions of SiH_4, Si, and N_2O to the total absorption signal. But after few tens of microseconds, the absorption was almost entirely due to oxygen atoms. Thus, the increase and decrease of the absorption level mainly corresponded to the production and consumption of O atoms, respectively.

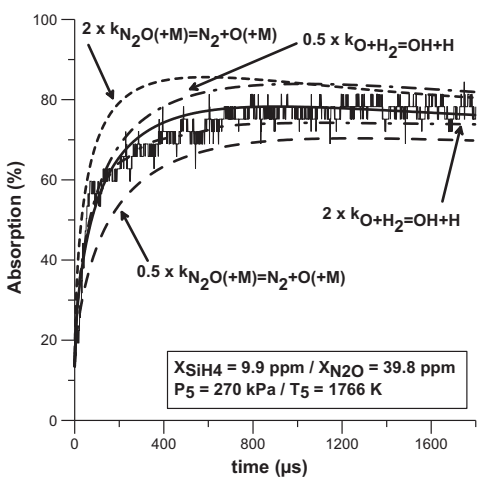

Fig. 3. Example of experimental and calculated absorption profiles obtained at 130.5 nm at low shock temperature for a SiH_4–N_2O–Ar mixture.

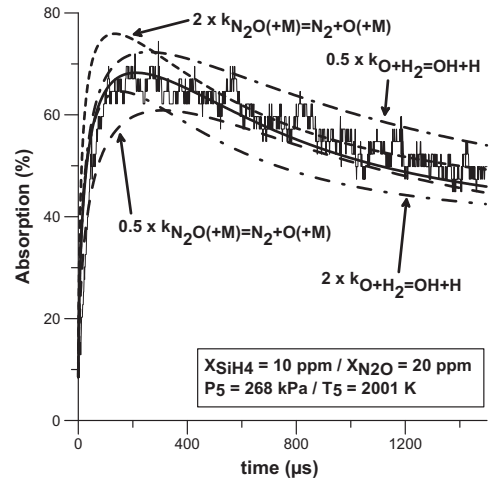

Fig. 4. Example of experimental and calculated absorption profiles obtained at 130.5 nm at shock intermediate temperature for a SiH_4–N_2O–Ar mixture.

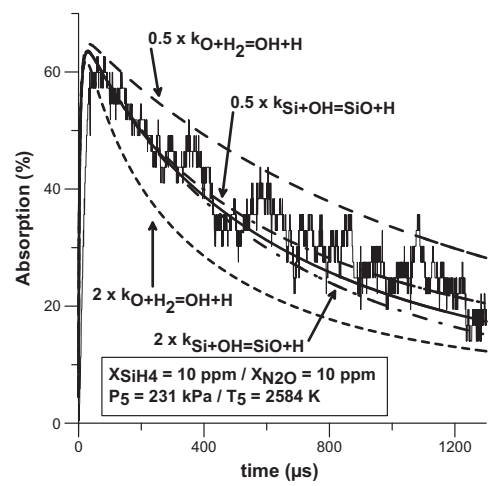

Fig. 5. Example of experimental and calculated absorption profiles obtained at 130.5 nm at high shock temperature for a SiH_4–N_2O–Ar mixture.

Figures 3–5 also compare the predictions of the developed kinetic scheme with the experimental absorption profiles. Although the contributions of SiH_4, Si and N_2O to the total absorption signal are rapidly negligible, they have been taken into account and the calculated absorption profiles were obtained from the following relationship:

$$A_{cal} = 1 - \exp(-6.91 \times 10^{-17} \times \ell \times [SiH_4] - 5.62 \times 10^{-17} \times \ell \times [Si] - 2.6 \times 10^{-17} \times \ell \times [N_2O] - 3.38 \times 10^{-10} \times \ell \times [O]^{0.625})$$

The time-dependant concentrations of SiH_4, Si, N_2O, and O were taken from the kinetic simulation. In the three cases presented (Figs. 3–5), the

agreement between the experimental and the calculated profiles is very satisfactory. For a total of 16 experiments performed, a good agreement was observed.

In order to underline the important elementary reactions for the kinetics of O atoms, first-order sensitivity analyses have been performed at temperature between 1700 and 2600 K. Calculations have been run over 2 ms and the sensitivity coefficient for a given reaction corresponds to the maximum value. Coefficients have been normalised with respect to the highest coefficient. The SiH_4 content was set at 10 ppmv and the N_2O content was decreased from 40 ppmv (at 1700 K) to 10 ppmv (at 2600 K). The pressure ranged from 608 kPa (at 1700 K) to 253 kPa (at 2600 K). Figure 6 summarizes the results obtained. Whatever the temperature is, the two most important reactions are: $N_2O(+M) = N_2 + O(+M)$ and $O + H_2 = OH + H$. As the temperature increases, the O atom consumption is faster: it justifies that the $O + H_2 = OH + H$ reaction becomes more sensitive than the $N_2O(+M) = N_2 + O(+M)$ reaction. At high temperatures, the $Si + OH = SiO + H$ and $OH+H_2 = H_2O + H$ reactions are also important. On the contrary, $H+O_2 = O + OH$ is important only at low temperatures. Finally, the $Si + N_2O = SiO + N_2$ reaction is sensitive only around 2300 K. The effect of multiplying the rate constants of the most important reactions by a factor of 2 and 0.5 on the absorption profiles is illustrated in Figs. 3–5. At low and intermediate temperatures, the variation of the rate constants of the $N_2O(+M) = N_2 + O(+M)$ and $O + H_2 = OH + H$ reactions induces a shift of the calculated profiles from the experimental ones. At high temperatures, the variation of the rate constant of the $O + H_2 = OH + H$ reaction induces either a too fast or a too slow consumption of O atoms. The variation of the rate constant of the $Si + OH = SiO + H$ reaction involves only a small shift of the calculated profile but its influence increases with time.

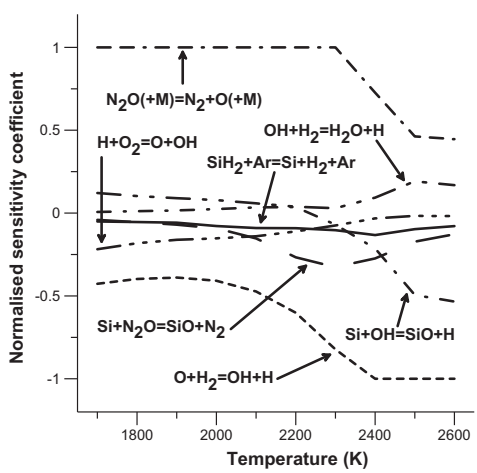

Fig. 6. Evolution of normalised sensitivity coefficients with respect to O atoms as a function of temperature. Conditions: $X_{SiH4} = 10$ ppmv; $X_{N_2O} = 10$–40 ppmv; $T = 1700$–2600 K; $P = 253$–608 kPa.

Figure 7 shows a schematic of a reaction pathway analysis performed at 2001 K and 268 kPa. It is seen that the successive decompositions of silane and silylene lead to the production of H_2 and Si. Silicon atoms mainly react with N_2O through the $Si + N_2O = SiO + N_2$ reaction. Small fractions of Si atoms also react either with N_2O to form $SiN + NO$ or with OH radicals to form $SiO + H$. Although a significant fraction of N_2O reacts with Si, its decomposition is the dominant pathway. The O atoms thus produced react almost entirely with H_2 to form OH and H. The totality of H_2O is produced by the $OH + H_2 = H_2O + H$ reaction. The other reaction products, SiO(s) and SiO_2(s), are mainly formed by $SiO + SiO = SiO(s) + SiO(s)$ and by $SiO + OH = H + SiO_2(s)$, respectively.

From the sensitivity and the reaction pathway analyses, it can be concluded that the dynamics of O atoms in SiH_4–N_2O–Ar mixtures are dominated by the $N_2O(+M) = N_2 + O(+M)$ and $O + H_2 = OH + H$ reactions. As illustrated in Figs. 3–5, the uncertainties associated with the rate constants of these reactions can induce a poor accuracy of the simulated absorption profiles. However, the rate constants included in the present kinetic scheme for these reactions came from two relatively recent studies performed in our laboratory [9,11] and are known within 30% of uncertainty, assuring an accurate prediction of the O-atom concentrations. The reactions between Si and N_2O result in a limited production of O atoms by N_2O thermal decomposition. Although these reactions are not sensitive, their inclusion into the kinetic scheme is necessary to obtain a good agreement with the experimental profiles.

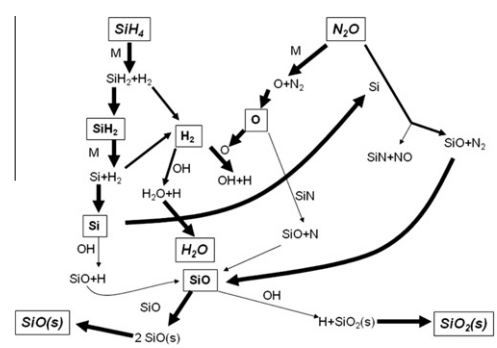

Fig. 7. Schematic of a reaction pathway analysis performed at 2001 K and 268 kPa for a SiH_4–N_2O–Ar mixture. Composition: $X_{SiH4} = 10$ ppmv; $X_{N_2O} = 20$ ppmv.

To check the validity of the developed model, previous shock tube data obtained with SiH_4–N_2O–Ar mixtures [8] have also been chosen for comparison.

Figure 8 compares the predictions of the mechanism with the experimental Si and N profiles from reference [8]. In both cases, there is a good agreement between the calculated and the experimental profiles. Figure 8 also illustrates the effect on the concentration profiles of varying the rate constant of the most important reactions. The Si profile is mainly controlled by the $Si + N_2O = SiO + N_2$ reaction. The most important reaction for the N profile is $Si + N_2O = SiN + NO$. The following reactions are also quite sensitive for the N profile: $SiN + O = SiO + N$; $N_2O(+M) = N_2 + O(+M)$; $Si + NO = SiO + N$ and $SiH_2 + Ar = Si + H_2 + Ar$. From modeling and sensitivity studies, it can be concluded that the kinetics of Si and N atoms in the experimental conditions used in reference [8] are mainly controlled by the reactions between Si and N_2O. As previously underlined, the rate constant of these competitive reactions have been determined by Mick and Roth using the ARAS technique applied to Si- and N-atoms [8]. These rate constants are known within 50% of uncertainty.

3.3. Oxidation of silane by nitric oxide and oxygen

In order to extend the validity of the kinetic model, previous shock tube data obtained with SiH_4–NO–Ar mixtures [20] and with SiH_4–O_2–Ar mixtures [19] have been taken for comparison.

Figure 9 shows the results obtained by modeling Si, N and O absorption and concentration profiles for SiH_4–NO–Ar mixtures in the experimental conditions of reference [20]. In each case, a satisfactory agreement can be noted. The effect on the profiles of varying the rate constant of the most important reactions is also shown in Fig. 9. The experimental profiles are quite sensitive to the rate constants of the $Si + NO = SiO + N$ and $NO + N => N_2 + O$ reactions. For the first reaction, the rate coefficients were obtained by [20] using the first-order method and is known within 40% of uncertainty. For the second reaction, it was necessary to include the rate parameter recommended by Baulch et al. [14] in order to obtain a satisfactory agreement for the N profile. In the original model [13], the rate constant of this reaction was calculated from the reverse reaction and the equilibrium constants, leading to poor predictions.

Figure 10 shows the results obtained by modeling Si and O concentration profiles for SiH_4–O_2–

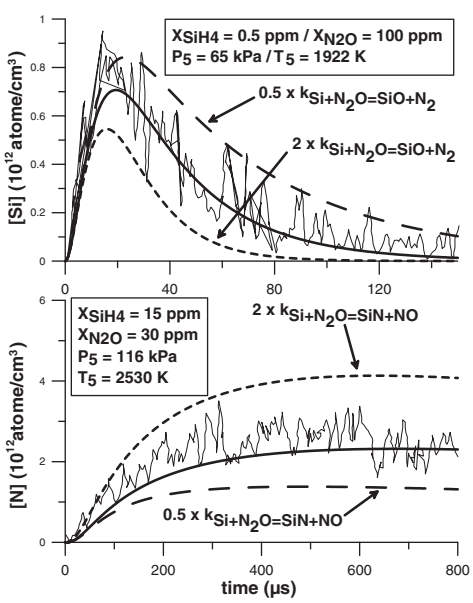

Fig. 8. Example of Si and N concentration profiles obtained at 251.6 and 119.9 nm, respectively, for SiH_4–N_2O–Ar mixtures. Experiments taken from reference [8] and calculated from this work.

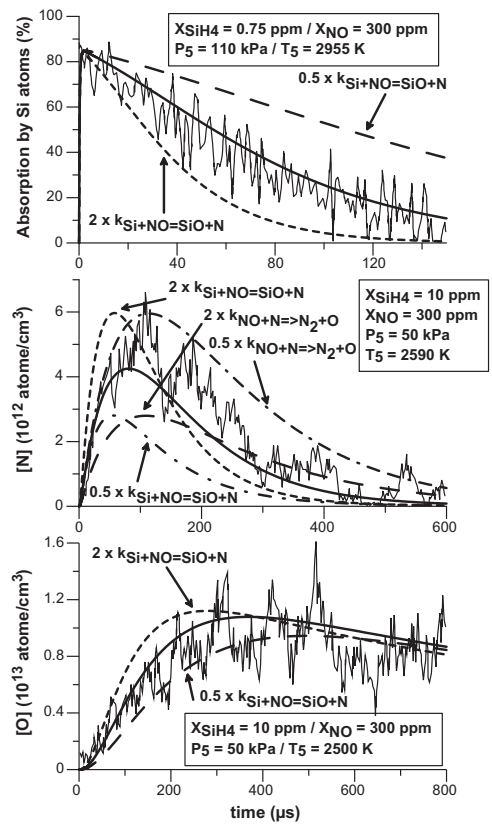

Fig. 9. Example of Si, N and O absorption and concentration profiles obtained at 251.6, 119.9, and 130.5 nm, respectively, for SiH_4–NO–Ar mixtures. Experiments taken from reference [20] and calculated from this work.

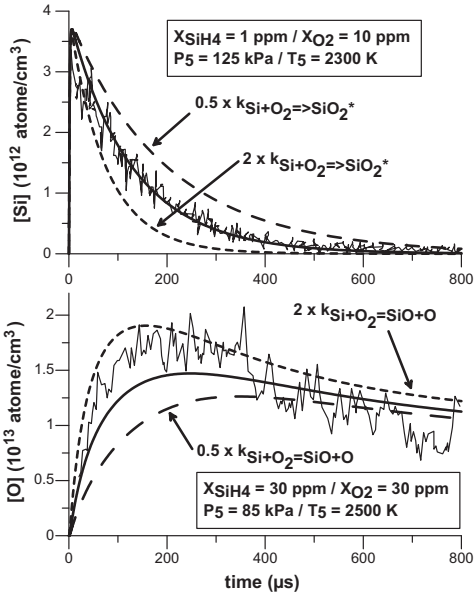

Fig. 10. Example of Si and O concentration profiles obtained at 251.6 and 130.5 nm, respectively, for SiH_4–O_2–Ar mixtures. Experiments taken from reference [19] and calculated from this work.

Ar mixtures in the experimental conditions of reference [19]. Again, a satisfactory agreement can be observed. The effect on the profiles of varying the rate constant of the most important reactions is also shown in Fig. 10. The Si and N profiles are sensitive to the reactions between Si and O_2. For these reactions Mick proposed a complex mechanism that includes the following reactions: $Si + O_2 = SiO + O$; $Si + O_2 = SiO_2^*$; $SiO_2^* + Ar = SiO_2 + Ar$ and $SiO_2^* + Ar = SiO + O + Ar$. Very few data are available in the literature to discuss about these kinetic paths; experiments are underway in our laboratory to obtain additional data.

4. Conclusion

In the present study, the kinetics of O atoms in SiH_4–N_2O–Ar mixtures behind reflected shock waves have been investigated by ARAS, in the range $T_5 = 1725$–2584 K and P_5=231–601 kPa. Absorption cross sections of silane and silicon atoms have been determined in order to take into account their contribution to the absorption at the O-triplet emission line. A chemical kinetic model has been developed and validated against the present data. Sensitivity and reaction pathway analyses have shown that the kinetics of O atoms were dominated by the two following reactions: $N_2O(+M) = N_2 + O(+M)$ and $O + H_2 = OH + H$. Previous experimental results from the literature for the oxidation of silane by nitrous oxide, nitric oxide and oxygen behind reflected shock waves have also been addressed. In these specific conditions, the most important reactions were found to be $Si + N_2O = SiO + N_2$, $Si + N_2O = SiN + NO$, $Si + NO = SiO + N$, $NO + N => N_2 + O$ and $Si + O_2 =$ products. Most of these elementary reactions have been already studied and their rate constants are known with a good accuracy. However, numerous additional experimental data would be necessary in order to check the validity of these rate coefficients and to allow a precise description of the dynamics of silane-based mixtures. Various experimental programs are on-going in our laboratory in order to obtain these new data.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.076.

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